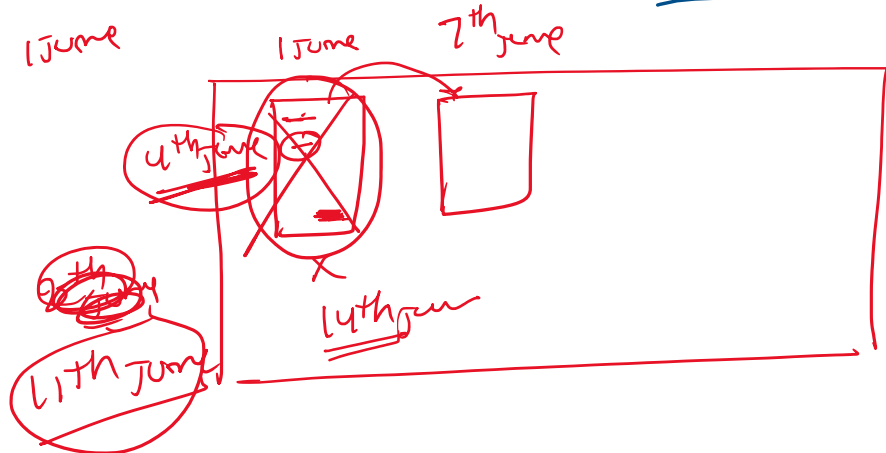
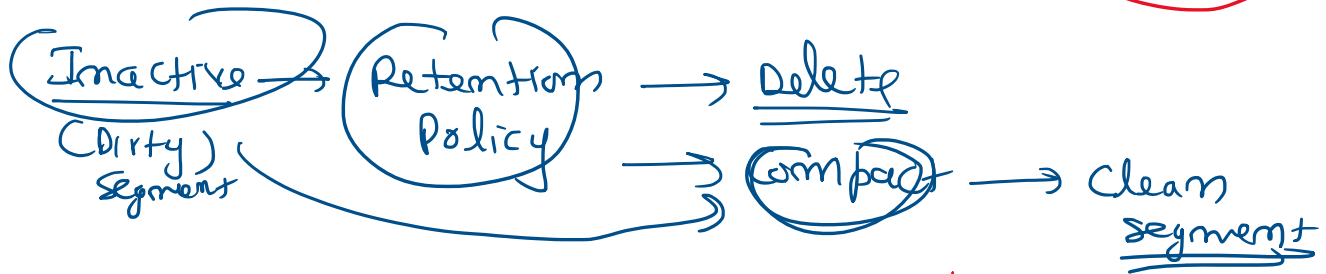
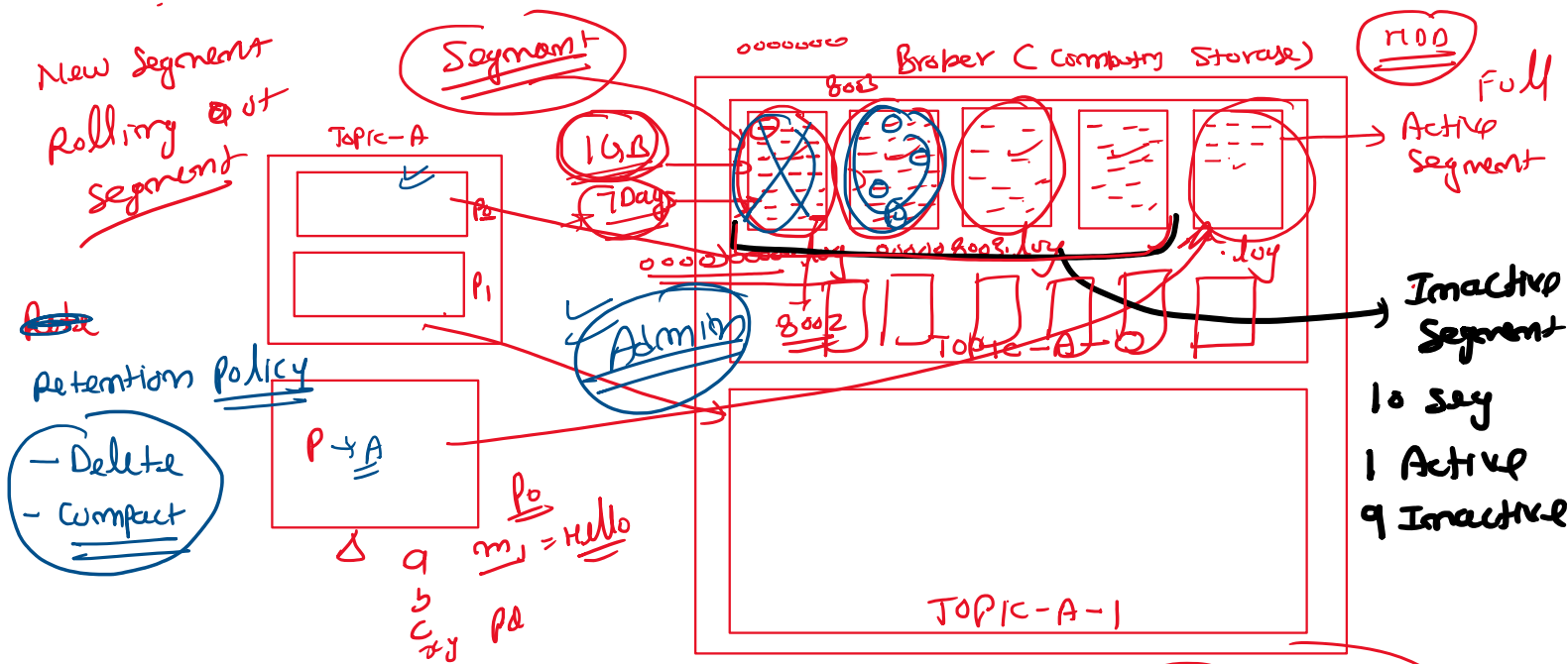




Partitioner

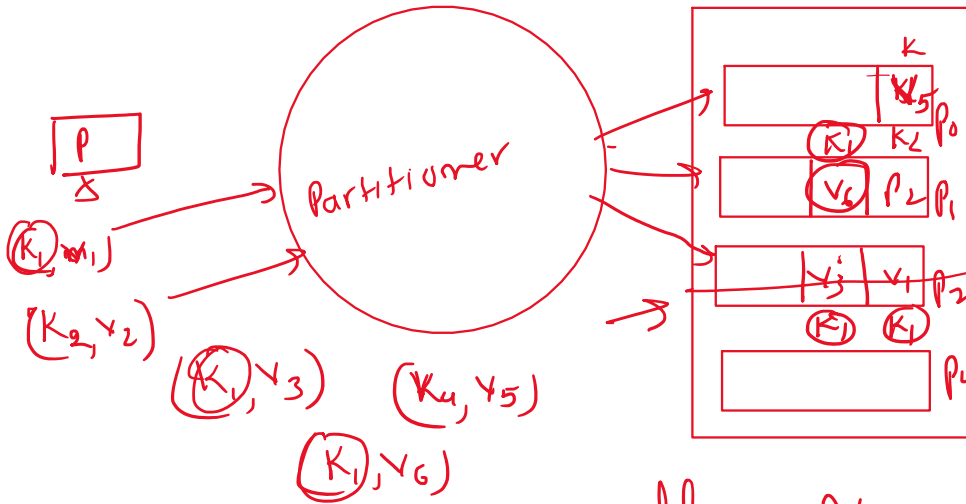
Default Partition

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99

Partitioner

Default Partition

$$\text{Partition Index (PI)} = \text{Hash}(\text{key}) \% \text{No. of Partitions}$$



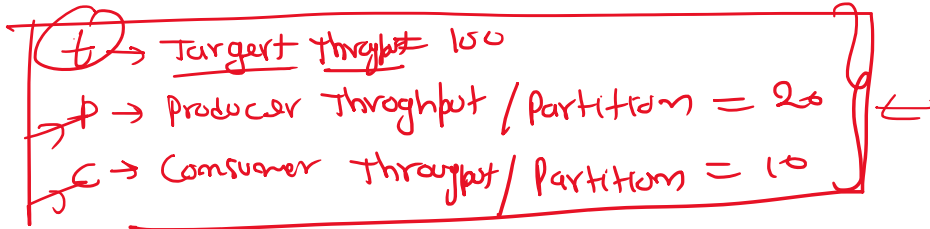
$$PI = \text{Hash}(K_1) \% 3 = 4 \% 3 = 1 \rightarrow P_1$$

$$PI = \text{Hash}(K_2) \% 3 = 400 \% 3 = 1 \rightarrow P_1$$

$$PI = \text{Hash}(K_4) \% 3 = 27 \% 3 = 0 \rightarrow P_0$$

$$PI = \text{Hash}(K_1) \% 4 = 4 \% 4 = 0 \rightarrow P_0$$

well in Advance $\rightarrow$ Good No. of Partitions



$$\begin{aligned} \text{No. of Partitions} &= \max\left(\frac{t}{p}, \frac{t}{c}\right) \\ &= \max\left(\frac{100}{20}, \frac{100}{10}\right) = \max(5, 10) \\ &= \underline{\underline{10}} \end{aligned}$$

4000 / Broker

200,000 / Cluster

